

DEEP LEARNING FOR FINANCE

Project # 1

Scenario

Suppose you spearhead the optimization engineering team for a statistical arbitrage division. As your first responsibility, you will be tasked with finding optimal configurations of parameters in models that trade the firm's capital. This is a very important—and highly visible—role. Perform it correctly and your group's expertise will result in clear value added. Perform it incorrectly and...well, let's not do that.

Tasks

Step 1

- a. Gather information on the time series of the prices (i.e. the levels) of any security of your choice: equity, cryptocurrencies, options, bonds, volatilities... Provide a characterization of the main properties of that time series: summary statistics about its distribution, persistence, stationarity...
- b. Perform the same analysis for a transformed version of the time series that is stationary (e.g. taking first or second differences, taking logs...). To provide arguments about stationarity, you can, for instance, generate visualizations of the time series and use the Augmented Dickey-Fuller (ADF test).
- c. Use [fractional differencing](#) to obtain a new version of the time series that is stationary but retains relevant information about the data. As above, provide a characterization of the main properties of that time series.
- d. As a group, be sure to comment on the 3 representations of the data.

Note: The time series should not exceed 2,000 observations.

Note: data need not be daily data if you have access to intra-day data you can use (e.g. cryptocurrency data).

Step 2

Your team has been requested to develop a Deep Learning model that allows you to predict the time series you have analyzed in Step 1. First, you are going to consider a Multi-Layer Perceptron model (MLP or Artificial Neural Network).

- a. Using past information of the time series in levels as inputs, build and train an MLP that is designed to predict the level of the time series.
- b. Using past information of the stationary time series as inputs, build and train an MLP that is designed to predict the stationary version of the time series.
- c. Using past information of the fractionally differenced time series as inputs, build and train an MLP that is designed to predict the fractionally-differenced version of the time series.
- d. As a group, evaluate the prediction performance of the three models. Provide arguments on why some models perform better than others.

Note: “past information” refers to including several lags of the time series, feel free to include as much or as little information as you wish. You can set the models as regression or classification problems (or both).

Step 3

Now you are going to compare the performance of the MLPs with Convolutional Neural Network (CNN) architectures. For that, you are going to transform each of the three versions of the time series you have constructed above and obtain their image representation using Gramian Angular Field (GAF).

- a. Using the GAF representation of the time series in levels, build and train a CNN that is designed to predict the level of the time series.
- b. Using the GAF representation of the stationary time series, build and train an MLP that is designed to predict the stationary version of the time series.
- c. Using the GAF representation of the fractionally-differenced time series, build and train a CNN that is designed to predict the fractionally-differenced version of the time series.
- d. As a group, evaluate the prediction performance of the three models. Provide arguments on why some models perform better than others.

Note: “past information” refers to including several lags of the time series, feel free to include as much or as little information as you wish. You can set the models as regression or classification problems (or both).

Step 4

Discuss the different results that you obtain between the CNN and MLP architectures. Provide your own educated guesses based on the different ways in which each architecture processes the data and based on the properties of the particular time series that is the focus of your analysis.

Step 5

As a group, please submit all files that reproduce the assignment. This will include the Python file for all the steps as well as work together on a report that contains the answers to all the questions above (except for the code) in a clear and organized manner (think about how you would present this to your boss).